

R E V I E W

Obesity and ceramides: The Web of Science and Scopus bibliometric analysis (2004–2025)

RAIKUL KOSMURATOVA, YERLAN BAZARGALIYEV, KHATIMYA KUDABAYEVA, YESSET SAGYNAYEV

Department of Internal Diseases № 1, West Kazakhstan Marat Ospanov Medical University, Aktobe, Kazakhstan;

ABSTRACT

Background and aim: Research on ceramides has expanded rapidly over the past two decades, reflecting increasing recognition of their role in obesity-related metabolic disturbances and cardiometabolic risk. However, the structure, dynamics, and global distribution of this research field have not been systematically evaluated. This study aimed to perform a comprehensive bibliometric analysis of human studies investigating the association between obesity and ceramides from 2004 to 2025.

Methods: A bibliometric analysis was conducted using publications retrieved from the Scopus and Web of Science Core Collection databases. Original human research articles published in English between 2004 and 2025 were included. After screening and removal of duplicates, 173 publications were analyzed. Bibliometric indicators included annual publication trends, citation metrics, productive journals, authors, institutions, countries, international collaboration networks, and keyword co-occurrence patterns. Analyses and visualizations were performed using Bibliometrix (R), Biblioshiny, and complementary visualization tools.

Results: The 173 included publications originated from 97 sources and demonstrated a mean annual growth rate of 14.76%, with an average of 46.2 citations per document. Scientific output increased substantially after 2015, reaching a peak in 2024. Authorship analysis revealed a highly collaborative research environment, with an average of 10.6 authors per article and frequent international collaborations. Publications were concentrated in high-impact endocrinology and metabolism journals. The United States led in both publication volume and citation impact, while several European and Asia-Pacific countries showed high citation efficiency relative to



Received: 5 February 2026 | Accepted: 6 March 2026

Correspondence: Yesset Sagynayev / Department of Internal Diseases № 1, West Kazakhstan Marat Ospanov Medical University, Maresyev Street 68 030012 Aktobe, Kazakhstan / E-mail: eset.200104@gmail.com
ORCID: 0009-0006-3717-8402

output. Keyword trend analysis indicated a shift from general lipid metabolism and insulin resistance toward ceramides, lipidomics, and cardiovascular risk in recent years.

Conclusions: Research on obesity and ceramides has evolved into a mature, collaborative, and clinically oriented field with growing translational relevance. The increasing emphasis on lipidomics and cardiometabolic outcomes underscores the potential of ceramides as biomarkers and therapeutic targets in obesity-related disorders (www.actabiomedica.it)

Key words: obesity, ceramides, bibliometric analysis, lipidomics, insulin resistance

Introduction

Obesity remains one of the most challenging global health problems of the twenty-first century. According to World Health Organization, in 2022, 2.5 billion adults (18 years and older) were overweight. The World Obesity Federation's 2025 Atlas projects that the total number of adults living with obesity will increase by more than 115% between 2010 and 2030, from 524 million to 1.13 billion (1). Obesity is associated with increased risk of several diseases, including cardiovascular (2), diabetes mellitus (3) and cancer (4). There is an increasing interest in novel biomarkers, such as adipokines (5,6), cytokines (7) and "omics" biomarkers (8) implicated in obesity. Within the broad spectrum of biomarkers implicated in obesity, ceramides stand out as a rapidly expanding focus of research. Ceramides are bioactive lipids that become elevated in obesity and related cardiometabolic disorders (9). They contribute to obesity-driven inflammation by activating pathways such as TLR4 and NLRP3, whereas inhibition of ceramide synthesis has been shown to attenuate inflammatory responses (10).

Among the molecular signatures linked to these disturbances, ceramides have emerged as key lipid mediators that influence inflammatory pathways (10), and whole-body metabolic homeostasis (11). Over the past decade, research on ceramides has accelerated, driven by new analytical technologies, expanded lipidomics platforms, and growing recognition that ceramide species may represent both biomarkers (12,13) and potential therapeutic targets in obesity-related disorders.

Despite this surge in interest, the field has become increasingly complex. Studies span molecular biochemistry (14–16), adipose tissue biology (17), endocrinology (18), hepatic steatosis (19), gut dysbiosis (20), pharmacology. In such a rapidly evolving and multidisciplinary research environment, it becomes increasingly difficult to track how scientific priorities shift over time. Bibliometric analysis provides a structured and quantitative approach to address these questions by mapping publication trends, citation networks, collaborative structures and thematic evolution within a research field. This approach enables identification of influential contributors, emerging research hotspots and translational trajectories that are not readily apparent from narrative reviews alone.

Methods

Search strategy

A comprehensive bibliometric search was conducted using two major international scientific databases: Scopus (Elsevier) and Web of Science Core Collection (Clarivate Analytics). We collected and analyzed data from 173 publications indexed in the Web of Science and Scopus databases, covering the period from 2004 to 2025. Although isolated studies on ceramides were published earlier, 2004 marked the beginning of sustained human-based clinical research on ceramide involvement in obesity. Since this study focused exclusively on human research to ensure clinical relevance, 2004 was selected as the starting point.

No additional grey literature sources were included, as the study focused exclusively on peer-reviewed journal publications to ensure data reliability and consistency. The search strategy was designed to identify original research articles addressing the relationship between obesity and ceramides. To enhance search precision, Boolean operators were applied, along with Medical Subject Headings (MeSH) terms (<https://www.nlm.nih.gov/mesh/meshhome.html>), which enabled the construction of a precise search formula and ensured the maximum relevance of the results (Tables 1,2).

Study selection

Studies were included if they met the following criteria: (1) original research articles; (2) focused on obesity and ceramides as a primary or substantial research topic; (3) published in peer-reviewed journals; and (4) written in English. Articles were excluded if they were review articles, conference papers, editorials, letters, or other non-original publications, or if they were not relevant to the study topic based on title, abstract, or full-text evaluation. Review articles were excluded to avoid citation inflation and to ensure that the analysis reflected original empirical contributions rather than secondary syntheses of existing evidence.

Data extraction

A two-stage screening process was applied to select articles according to the predefined inclusion criteria. In the first stage, two authors independently screened the titles and abstracts of all records identified through the search strategy. Articles approved by both authors were subsequently assessed in full text during the second stage. Discrepancies were resolved through discussion until consensus was reached. All eligible publications

were exported in plain text format from Web of Science Core Collection and BibTeX format from Scopus, including bibliographic information, author keywords, citation data, and affiliation details. Duplicate records were identified and removed after merging the datasets. The datasets retrieved from Scopus and Web of Science were merged using a custom code in RStudio, and the combined dataset was exported in Microsoft Excel (xlsx) format for further analysis. Content and bibliometric analyses included the annual distribution of publications; contributions by countries, institutions, journals, and authors; keyword co-occurrence patterns; and temporal changes in keyword frequency. These analyses were used to identify research productivity, collaboration patterns, and emerging research topics related to obesity and ceramides (Figure 1).

Performance analysis

For analysis and visualization, we utilized Bibliometrix R (<http://www.bibliometrix.org>) in RStudio v.4.3.2 (Posit Software, PBC, Boston, Massachusetts, USA), Biblioshiny, and Flourish.studio (<https://flourish.studio/>)

Results

Summary of the papers

A total of 173 documents related to obesity and ceramides were identified from 97 sources, covering the period from 2004 to 2025. The field showed a mean annual growth rate of 14.76%. The average age of publications was 6.9 years, and the mean number of citations per document was 46.2. Authorship analysis showed a highly collaborative research environment,

Table 1. Query for Scopus.

Query
(TITLE-ABS-KEY "Obesity, Abdominal" OR "Abdominal Obesity" OR "Central Obesity" OR "Visceral Obesity" OR "Obesity, Metabolically Benign" OR "Metabolically Healthy Obesity" OR "Metabolically Benign Obesity" OR "Obesity, Morbid" OR "Morbid Obesity" OR "Severe Obesity" OR "Overweight" OR "Obesity" "Adult-Onset Diabetes Mellitus" OR "Diabetes Mellitus, Adult Onset") AND "Ceramides" OR "Ceramide" OR "Cerebrosides" OR "Galactosylceramides" OR "Glucosylceramides" OR "Globosides" OR "Lactosylceramides" OR "Trihexosylceramides" AND (LIMIT-TO (LANGUAGE, "English"))

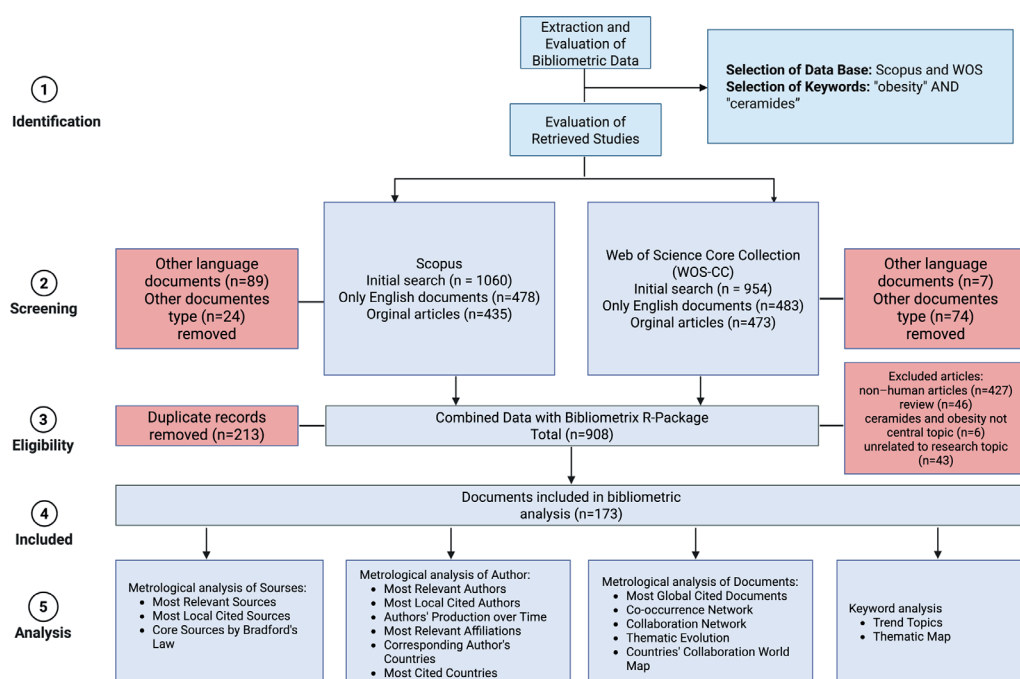


Figure 1. Search process.

with an average of 10.6 co-authors per article and 23.1% of publications involving international collaboration. Only one single-authored publication was identified.

Annual analysis of publications

Temporal analysis revealed a clear shift from limited early research activity to sustained growth in scientific output. The number of publications remained low and sporadic from 2004 to 2014, fluctuating between 0 and 5 articles per year. Beginning in 2015, a marked increase in publication activity was observed, with 12 articles published in that year. From 2016 to 2020, the number of publications remained relatively stable, ranging from 6 to 11 articles annually. A substantial rise was evident from 2021 onward, with a pronounced increase in publication output, reaching 15 articles in 2021 and peaking in 2024 with 23 publications. High publication rates were also observed in 2022 (21 articles). In 2025, a slight decline was noted, with 18 publications, although the overall level remained considerably higher than in earlier years (Figure 2).

Source analysis

Publications on obesity and ceramides are concentrated in a limited number of endocrinology and metabolic journals, with DIABETES and OBESITY being the most prominent, followed by the Journal of Clinical Endocrinology and Metabolism and Diabetologia (Table 3). The volume of publications and the total number of citations are distributed unevenly among the journals. Most publications belong to the first quartile (Q1) of the Journal Citation Reports, including DIABETES, OBESITY, Journal of Clinical Endocrinology and Metabolism, Diabetologia, Lipids in Health and Disease, American Journal of Physiology – Endocrinology and Metabolism, International Journal of Obesity, and JCI Insight, while Metabolites belongs to the second quartile (Q2). Leading journals are published by professional scientific societies and international publishers, including the American Diabetes Association, Endocrine Society, American Physiological Society, American Society for Clinical Investigation, Springer Nature, Springer Science and Business Media, John Wiley & Sons, BioMed Central, and Multidisciplinary Digital Publishing Institute (MDPI).

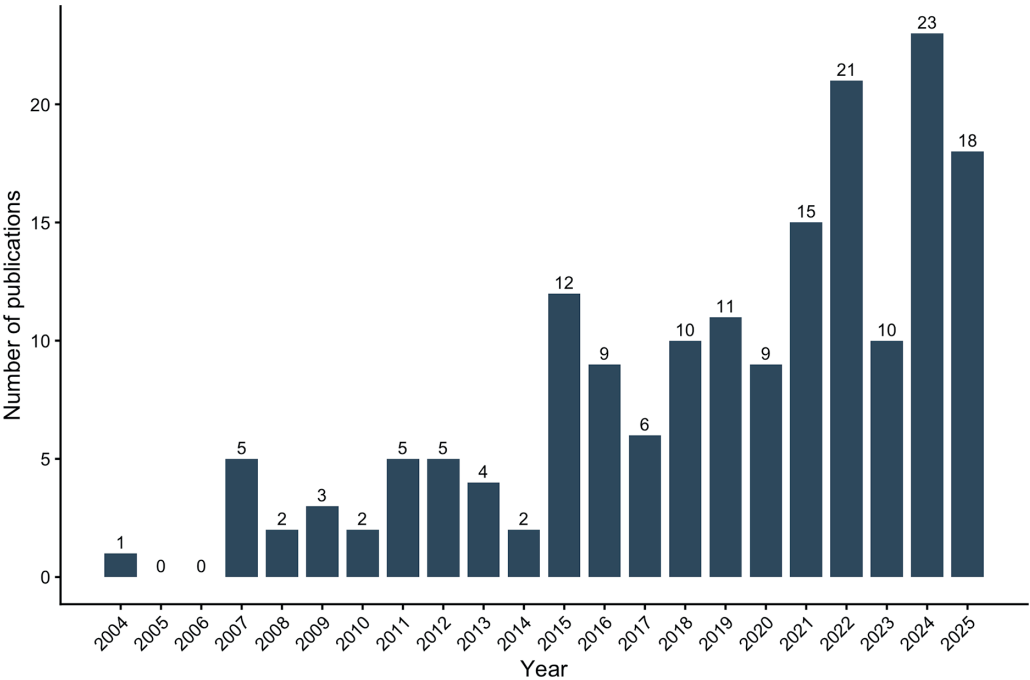


Figure 2. Annual publication trends over time (2004-2025).

Table 2. Query for Web of Science database.

Nº	Queries
#1	“Obesity, Abdominal” OR “Abdominal Obesity” OR “Central Obesity” OR “Visceral Obesity” OR “Obesity, Metabolically Benign” OR “Metabolically Healthy Obesity” OR “Metabolically Benign Obesity” OR “Obesity, Morbid” OR “Morbid Obesity” OR “Severe Obesity” OR “Overweight” OR “Obesity” “Adult-Onset Diabetes Mellitus” OR “Diabetes Mellitus, Adult Onset” (Title) or “Obesity, Abdominal” OR “Abdominal Obesity” OR “Central Obesity” OR “Visceral Obesity” OR “Obesity, Metabolically Benign” OR “Metabolically Healthy Obesity” OR “Metabolically Benign Obesity” OR “Obesity, Morbid” OR “Morbid Obesity” OR “Severe Obesity” OR “Overweight” OR “Obesity” “Adult-Onset Diabetes Mellitus” OR “Diabetes Mellitus, Adult Onset” (Author Keywords) or “Obesity, Abdominal” OR “Abdominal Obesity” OR “Central Obesity” OR “Visceral Obesity” OR “Obesity, Metabolically Benign” OR “Metabolically Healthy Obesity” OR “Metabolically Benign Obesity” OR “Obesity, Morbid” OR “Morbid Obesity” OR “Severe Obesity” OR “Overweight” OR “Obesity” “Adult-Onset Diabetes Mellitus” OR “Diabetes Mellitus, Adult Onset” (Abstract)
#2	“Ceramides” OR “Ceramide” OR “Cerebrosides” OR “Galactosylceramides” OR “Glucosylceramides” OR “Globosides” OR “Lactosylceramides” OR “Trihexosylceramides” (Title) “Ceramides” OR “Ceramide” OR “Cerebrosides” OR “Galactosylceramides” OR “Glucosylceramides” OR “Globosides” OR “Lactosylceramides” OR “Trihexosylceramides” (Author Keywords) “Ceramides” OR “Ceramide” OR “Cerebrosides” OR “Galactosylceramides” OR “Glucosylceramides” OR “Globosides” OR “Lactosylceramides” OR “Trihexosylceramides” (Abstract)
#3	#1 and #2

Author’s productivity

Analysis of authorship patterns indicates that a limited number of researchers account for a substantial proportion of publications in this field. Author

H-indices were retrieved from Scopus and reflect global citation metrics rather than dataset-restricted values. Peter Meikle was the most productive author with 11 publications, followed by Bryan Bergman and Leigh Perreault. Several authors demonstrated high citation

Table 3. Most relevant sources.

Sources	Articles	Total citation	JCR category (quartile)	Publisher
Diabetes	12	2404	Q1	American Diabetes Association Inc.
Obesity	12	592	Q1	John Wiley and Sons Inc
Journal of clinical endocrinology and metabolism	9	491	Q1	Endocrine Society
Diabetologia	6	690	Q1	Springer Science and Business Media Deutschland GmbH
Lipids in health and disease	6	128	Q1	BioMed Central Ltd
American journal of physiology - endocrinology and metabolism	5	189	Q1	American Physiological Society
Metabolites	5	52	Q2	Multidisciplinary Digital Publishing Institute (MDPI)
International journal of obesity	4	140	Q1	Springer Nature
Jci insight	4	299	Q1	American Society for Clinical Investigation

Table 4. Top 10 most prolific authors.

Authors	Articles	H-index	TC	Affiliation
Meikle, Peter	11	70	583	Baker Heart and Diabetes Institute, Melbourne, Australia
Bergman, Bryan	10	45	530	University of Colorado Anschutz Medical Campus, Aurora, USA
Perreault, Leigh	9	43	489	University of Colorado Anschutz Medical Campus, Aurora, USA
Brozinick, Joseph	6	42	342	Eli Lilly and Company, Indianapolis, USA
Chabowski, Adrian	6	47	159	Medical University of Bialystok, Bialystok, Poland
Goodpaster, Bret	5	104	1163	Translational Research Institute, AdventHealth, Orlando, USA
Jensen, Michael Dennis	5	92	122	Endocrine Research Unit, Mayo Clinic, Rochester, USA.
Siddall, Philip John	5	46	329	University of Sydney, Sydney, Australia
Amati, Francesca	4	29	1019	Université de Genève, Geneva, Switzerland
Baranowski, Marcin	4	30	324	Medical University of Bialystok, Bialystok, Poland

counts relative to publication number, including Bret Goodpaster, Michael Dennis Jensen and Francesca Amati. Author affiliations were predominantly linked to academic medical centers in the United States, Australia and Europe. In addition, several publications included authors affiliated with industry institutions, such as Eli Lilly and Company (Table 4).

Countries and affiliations

Country-level publication counts were derived from author affiliations; therefore, a single document

could be attributed to more than one country. Analysis of institutional affiliations demonstrates that research output was unevenly distributed across countries and institutions, with several countries represented by multiple productive academic centers (Table 5).

At the institutional level, the Medical University of Bialystok was the leading contributor, followed by Shanghai Jiao Tong University. In contrast, research output from the United States was distributed across several institutions rather than concentrated within a single center. Four U.S.-based institutions—University of Pittsburgh, University of Utah, University of Illinois

Table 5. Most relevant affiliations.

Affiliation	Articles	Country
Medical University Bialystok	32	Poland
Shanghai Jiao Tong University	19	China
University of Pittsburgh	13	USA
University of Utah	12	USA
University of Copenhagen	11	Denmark
Amsterdam UMC - University of Amsterdam	9	The Netherlands
Monash University	9	Australia
National University of Singapore	9	Singapore
University of Illinois	9	USA
Vanderbilt University Medical Center	9	USA

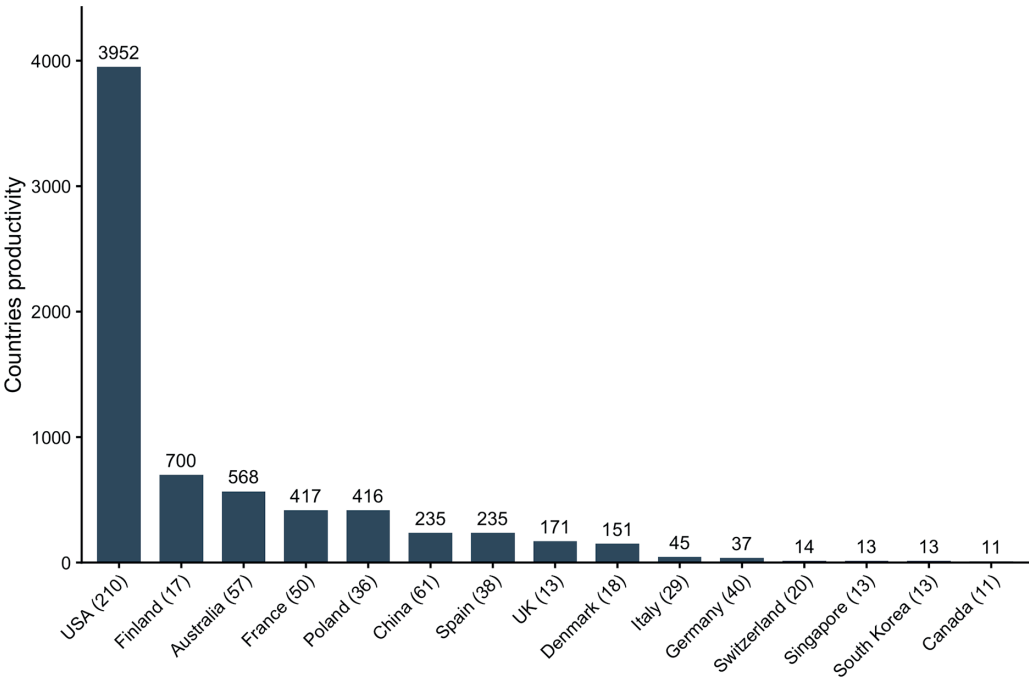


Figure 3. Country productivity.

and Vanderbilt University Medical Center-were among the most productive affiliations identified in this analysis. European institutions, including the University of Copenhagen and Amsterdam UMC – University of Amsterdam, were also represented among the most relevant affiliations. In addition, institutions from the Asia-Pacific region, such as Monash University and the National University of Singapore, contributed to the overall publication output, reflecting broad international participation in this research field.

An analysis of national scientific productivity revealed marked heterogeneity between the number of publications and the total number of citations. The United States dominates both in the number of publications ($n = 210$) and the total number of citations ($n = 3952$), significantly outperforming all other countries (Figure 3). This gap clearly persists when compared with both China and European countries, highlighting the asymmetric distribution of scientific influence.

China ranks second in the number of publications, but its total citations remain significantly lower than those of the United States. Thus, while publication activity is comparable, there is a difference in the accumulated citation impact. Australia and France demonstrate a consistent contribution to the literature, combining a moderately high volume of publications with an intermediate level of citation impact. Significant differences in the ratio of publications to citations are observed among European countries. Germany, with 40 published papers, has a relatively low total citation count, while Poland, with fewer publications, has accumulated a higher citation count. This difference persists both in direct comparisons between Germany and Poland and when Germany is compared with other European countries.

A separate group consists of countries with a limited number of publications but a high total citation count. Finland, with 17 publications, has accumulated 700 citations, representing one of the highest citation counts relative to the volume of scientific output. This contrast is particularly pronounced when comparing Finland with Germany and France, as well as with other countries with similar or even higher publication counts. A similar trend is observed in Denmark, the United Kingdom and Singapore, where relatively low publication volume is accompanied by significant citation impact.

In contrast, Switzerland, Italy, South Korea and Canada exhibit both low publication activity and a limited total citation count. This feature becomes particularly evident when comparing these countries with Finland, Denmark, and the United Kingdom.

A wide global network involving multiple countries was revealed during the analysis of international scientific collaboration. The strongest bilateral collaboration links were observed between the United Kingdom and New Zealand and between Australia and New Zealand, each showing the highest collaboration frequency. High-intensity collaborations were also identified between the United States and Japan, as well as China and Japan. Several countries acted as major collaboration hubs. South Korea demonstrated extensive international collaboration, with strong links to Australia, the United States, Switzerland, Finland and Italy. Similarly, the United States showed high

collaboration frequencies with Japan, Australia and South Korea, indicating its central role in the global research network (Figure 4). European countries such as Germany, the Netherlands, Switzerland, Finland and Italy participated actively in international collaborations, although with varying collaboration intensities. Cross-continental collaborations between North America, Europe, East Asia and Oceania were prominent, reflecting a high level of global connectivity in the research field.

Most relevant documents

The top three most cited articles on obesity and ceramides have significantly advanced the understanding of the relationship between ceramide metabolism and obesity (Table 6). In 2009, Haus et al. reported that plasma ceramide levels are elevated in obese subjects with type 2 diabetes and positively correlate with the severity of insulin resistance, receiving a total of 565 citations (31.39 citations per year) (21). Another highly cited study by Adams et al., published in 2004, demonstrated that ceramide content is increased in skeletal muscle from obese insulin-resistant humans, with 547 citations (22). In 2011, Amati et al. examined skeletal muscle triglycerides, diacylglycerols and ceramides in endurance-trained athletes and showed that insulin sensitivity can be preserved despite high intramuscular lipid content, while ceramide accumulation remains specifically associated with insulin resistance, with a total of 326 citations (23).

Trend topics

The temporal analysis of author keywords revealed distinct phases in the evolution of research themes related to obesity and ceramides (Figure 5). Only author keywords meeting a minimum occurrence threshold were included in the temporal analysis. During the early period, frequently occurring keywords included *triglyceride*, *adipose tissue*, *intraabdominal fat*, *fatty acids* and *protein kinase C*. These terms were present primarily within this time window and showed limited persistence in later years. In the subsequent period, keywords such as *insulin resistance*, *insulin sensitivity*, *adiponectin* and *lipids* appeared with increased frequency and broader

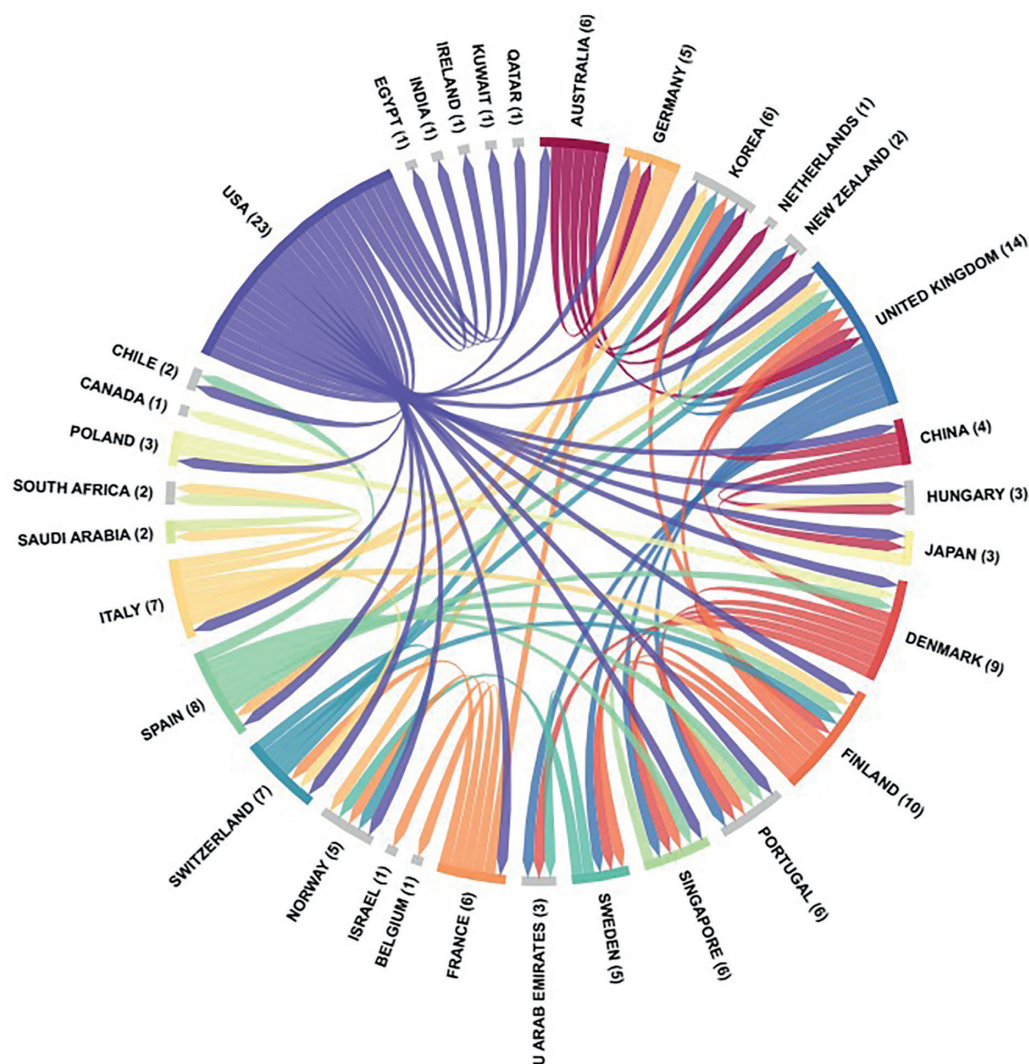


Figure 4. Countries collaboration.

temporal distribution, overlapping with both earlier and later phases. From 2018 onwards, the occurrence of keywords including *obesity*, *ceramide*, *sphingomyelin* and *lipidomics* increased markedly. Among these, obesity showed the highest frequency during the 2019–2020 period, while ceramide and lipidomics demonstrated sustained presence through the most recent years of the analysis. In the most recent period, additional keywords related to clinical conditions and outcomes, including cardiovascular disease, hypertension, dyslipidemia, risk factors, and biological marker, emerged and persisted through the end of the study period.

Discussion

In the present bibliometric analysis, we examined 173 publications focused on obesity and ceramides published between 2004 and 2025. The steady increase in scientific output, particularly after 2015, reflects the growing and sustained interest in ceramides as key bioactive lipids involved in obesity-related metabolic disturbances. Our analysis provides a comprehensive overview of publication dynamics, authorship patterns, collaboration networks and temporal trends within this research field.

Table 6. The most cited articles.

Rank	Article	1st author	DOI	Total Citations	Citation per Year	Year of Publication
1	Plasma ceramides are elevated in obese subjects with type 2 diabetes and correlate with the severity of insulin resistance	Jacob M. Haus	10.2337/db08-1228	565	31.39	2009
2	Ceramide Content Is Increased in Skeletal Muscle From Obese Insulin-Resistant Humans	John M. Adams	10.2337/diabetes.53.1.25	547	23.78	2004
3	Skeletal Muscle Triglycerides, Diacylglycerols, and Ceramides in Insulin Resistance: Another Paradox in Endurance-Trained Athletes?	Francesca Amati	10.2337/db10-1221	326	20.38	2011
4	Role of diacylglycerol activation of PKC θ in lipid-induced muscle insulin resistance in humans	Julia Szendroedi	10.1073/pnas.1409229111	325	25.00	2014
5	Exercise-induced alterations in intramyocellular lipids and insulin resistance: the athlete's paradox revisited	John J. Dubé	10.1152/ajpendo.00769.2007	296	15.58	2008
6	Circulating Ceramides Predict Cardiovascular Outcomes in the Population-Based FINRISK 2002 Cohort	Aki S. Havulinna	10.1161/ATVBAHA.116.307497	281	25.55	2016
7	Adipose Tissue Inflammation and Increased Ceramide Content Characterize Subjects With High Liver Fat Content Independent of Obesity	Maria Kolak	10.2337/db07-0111	262	13.10	2007
8	Plasma Lipid Profiling Shows Similar Associations with Prediabetes and Type 2 Diabetes	Peter J. Meikle	10.1371/journal.pone.0074341	249	17.79	2013
9	Effects of weight loss and exercise on insulin resistance, and intramyocellular triacylglycerol, diacylglycerol and ceramide	John J. Dubé	10.1007/s00125-011-2065-0	206	12.88	2011
10	Overeating Saturated Fat Promotes Fatty Liver and Ceramides Compared With Polyunsaturated Fat: A Randomized Trial	Fredrik Rosqvist	10.1210/jc.2019-00160	192	24.00	2019

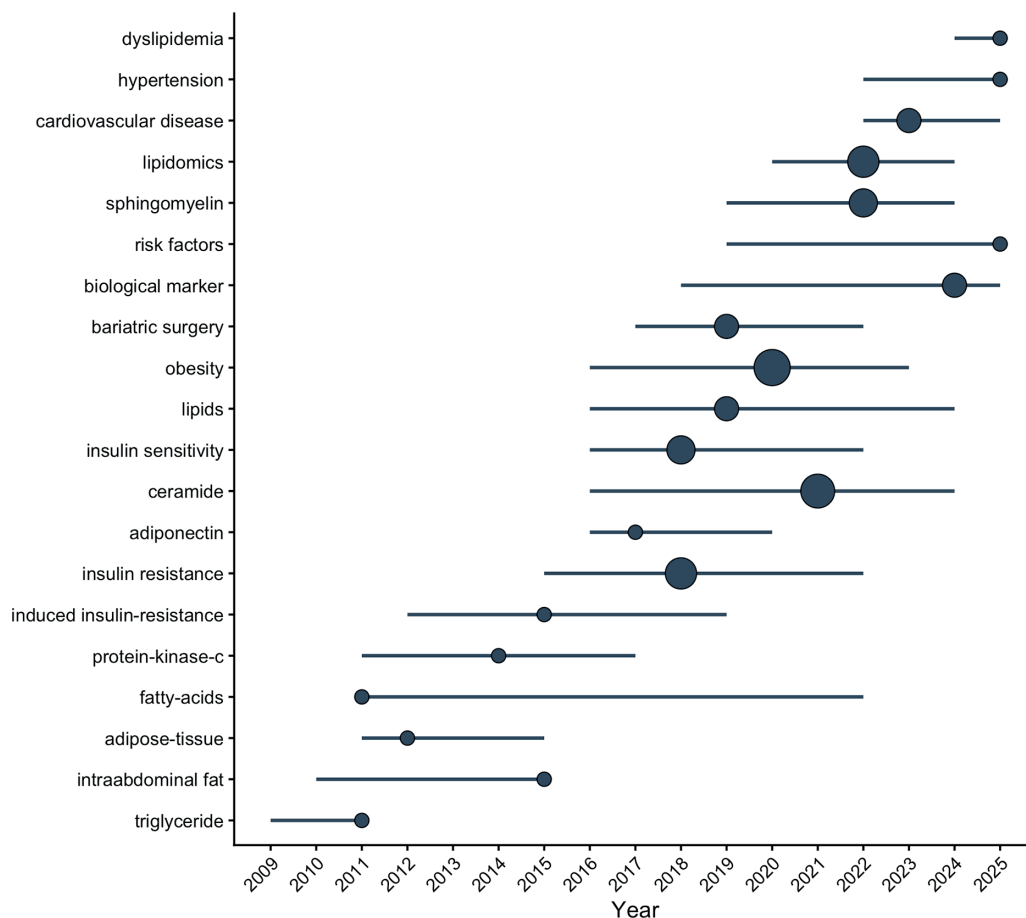


Figure 5. Trend topics.

The temporal distribution of publications revealed a prolonged period of limited research activity from 2004 to 2014, followed by a clear and sustained growth in output from 2015 onward. This shift likely reflects the increasing recognition of ceramides as central mediators linking excess adiposity with insulin resistance, chronic inflammation and cardiometabolic complications (24). The sharp rise in publications observed after 2021 further underscores the expanding role of lipidomics and advanced analytical techniques in obesity research, enabling more detailed characterization of sphingolipid metabolism and its clinical implications (25).

The relatively high mean annual growth rate and citation impact observed in this field indicate not only increasing productivity but also strong scientific influence. The average citation count of 46.2 per document suggests that studies addressing the role of ceramides in obesity

are widely referenced and contribute substantially to the broader metabolic research landscape (26). These findings highlight ceramides as a research focus of high relevance within endocrinology and metabolic medicine.

The concentration of publications within a limited set of high-impact, discipline-specific journals reflects the consolidation of research on obesity and ceramides within established endocrinology and metabolism venues. The dominance of journals such as *DIABETES*, *OBESITY*, and the *Journal of Clinical Endocrinology and Metabolism* suggests that ceramides are primarily investigated as mediators of insulin resistance, adipose tissue dysfunction and obesity-related metabolic complications, rather than as broader lipidomic or systems-biology biomarkers (26). Moreover, the uneven distribution of publication volume and citation impact across journals highlights differences in editorial scope

(27), audience reach and citation dynamics, which collectively shape the dissemination and recognition of scientific findings. At the same time, the clustering of output within diabetes-oriented journals implies that the scientific narrative is largely framed through insulin resistance and type 2 diabetes, potentially limiting broader integration into adjacent domains such as cardiovascular disease, hepatology and systemic inflammation. This disciplinary anchoring may shape both research priorities and citation dynamics.

The profile of top-cited authors suggests that ceramide research in obesity is driven by investigators embedded in broader metabolic and diabetes research programs rather than by a distinct, lipid-specialized community. Most highly cited contributors are affiliated with major translational and clinical research centers, including diabetes institutes, academic medical campuses and specialized metabolic research units, underscoring the clinical orientation of the field. The presence of pharmaceutical industry representation further indicates therapeutic interest in targeting ceramide pathways. Notably, citation impact does not strictly correlate with the number of publications on the topic, suggesting that conceptual or mechanistic landmark studies disproportionately shape the intellectual structure of the field (28).

At the institutional level, productivity appears concentrated within specific research hubs, including major U.S. academic centers and focused European institutions. Such clustering is consistent with established bibliometric evidence indicating that institutional concentration and research hub formation enhance visibility and citation impact. Geographically, the field demonstrates a stratified structure of scientific influence. The United States occupies a dominant position in both publication output and citation impact, suggesting not only quantitative leadership but also centrality in shaping the conceptual framework of ceramide-related metabolic research. This pattern may be explained by sustained federal investment in obesity and diabetes research through agencies such as NIH and NIDDK, which have historically prioritized metabolic disease (29). Bibliometric research consistently demonstrates that funding intensity correlates with increased citation performance at both institutional and national levels (30). China ranks second in publication output

yet does not exhibit proportional citation centrality. This divergence reflects a broader pattern observed in global science, where rapid quantitative expansion does not immediately translate into equivalent citation influence. Differences in network integration, journal positioning and participation in highly cited international collaborations may partially explain this gap (31). Within Europe, marked heterogeneity is evident. Poland demonstrates a relatively high citation impact despite a moderate publication volume, largely driven by concentrated activity within a single institutional hub. In contrast, Germany, while more prolific in output, shows comparatively lower citation accumulation in this field. Such discrepancies may reflect differences in research consolidation, publication strategies, or the presence of landmark studies that shape subsequent citation trajectories. The phenomenon aligns with the “Matthew effect” in science, whereby early high-impact contributions reinforce cumulative visibility advantages (32). A distinct group of high-efficiency contributors - including Finland, Denmark, the United Kingdom and Singapore - exhibits strong citation performance relative to publication volume. Smaller yet highly networked research systems often achieve elevated impact through participation in international consortia, publication in high-impact journals, and integration into collaborative research networks (33). Evidence from scientometric research indicates that internationally co-authored publications and larger collaborative teams tend to receive significantly more citations (34). Such mechanisms may underlie the observed citation efficiency of these countries.

Clinical lipidomics employs mass spectrometry and complex analytical workflows to profile hundreds of lipid species in human samples. This necessitates international collaboration to harmonize sampling, analytical protocols, and data interpretation, and to ensure that findings can be validated and translated into reliable clinical biomarkers (35). Prior analyses have consistently shown that internationally co-authored publications tend to achieve higher visibility and citation rates (36). Reviews focusing on ceramides as emerging biomarkers of lipotoxicity and metabolic dysfunction illustrate how such integrative approaches can enhance visibility and citation uptake, particularly when connecting lipid biology with clinical phenotypes

of obesity and metabolic syndrome (37). Countries such as the United States, East Asian nations and Australia emerge as key hubs in ceramide–obesity research likely because they combine strong academic traditions in metabolic disease. These regions host well-established research centres and consortia with expertise in diabetes, obesity and lipidomics (38). Moreover, national research programmes addressing the rising public health burden of metabolic disorders provide structural incentives for international cooperation and data sharing (39). Regional clusters in international research collaborations likely reflect long-standing academic ties, aligned research priorities, and eased ethical and regulatory coordination within connected regions. As a result, clusters such as East Asia, Europe and Oceania emerge not only because of scientific interest but also because of these structural and procedural advantages that promote sustained multilateral research engagement (40). Smaller or more specialized research systems may therefore achieve disproportionately high citation impact by focusing on niche topics, methodological rigor or strategically positioned collaborations, whereas larger systems may prioritize volume-driven productivity. From this perspective, the findings suggest that scientific influence in this field is increasingly determined by qualitative and network-based factors rather than by sheer production capacity (41).

The temporal evolution of author keywords provides a complementary and revealing perspective on the shifting priorities within this research domain. The early phase, dominated by terms such as “triglyceride,” “fatty acids,” and “protein kinase C,” reflects a foundational period focused on broad lipid metabolism and initial signaling mechanisms (42,43). This phase logically preceded the emergence of the most-cited muscle-centric studies, setting the stage for investigating specific lipotoxic mediators. The subsequent period saw a consolidation around core metabolic concepts, with keywords like “insulin resistance,” “insulin sensitivity,” and “adiponectin” achieving prominence and broad temporal distribution. This shift mirrors the dominance of insulin resistance pathophysiology in the most influential papers from the 2000s and early 2010s (44), confirming this as the central, enduring theme that unified the field. A pivotal transition occurred from approximately 2018 onward, marked by a decisive terminological shift. The surge in frequency of “ceramide” and “lipidomics” (45),

alongside the sustained peak of “obesity,” indicates the maturation of the field. This represents a move from studying ceramides as one component within general lipid biology to recognizing them as a defining and discrete research subject in their own right, directly linked to the disease state of obesity. This period corresponds with the expansion of research from muscle to hepatic and systemic contexts, as seen in the broader literature. Finally, the most recent emergence and persistence of keywords such as “cardiovascular disease,” “risk factors,” and “biological marker” signal the field’s current translational phase. This directly reflects the findings from high-impact population studies (e.g., Havulinna et al., 2016) and underscores the successful journey of ceramides from a mechanistic player to a validated component of clinical cardiometabolic risk stratification (46).

Conclusion

This bibliometric analysis demonstrates the substantial evolution of research on obesity and ceramides from an emerging mechanistic topic to an established and translationally relevant domain within metabolic medicine. The collaborative and internationally networked character of the field further reflects the methodological complexity of lipidomic research and the shared global priority of addressing cardiometabolic disease. Future research should prioritize deeper interdisciplinary integration across endocrinology, cardiovascular medicine, hepatology and systems biology to broaden the conceptual framing beyond diabetes-centered paradigms. Overall, this analysis provides a structured overview of the global research architecture in obesity-ceramide studies, identifies strategic gaps in interdisciplinary expansion and methodological standardization and highlights the need for coordinated international efforts to accelerate clinical translation and improve cardiometabolic risk stratification.

Ethic Approval: Ethical approval was not required because the study used only publicly available bibliometric data and did not involve human participants or identifiable personal information.

Conflict of Interest: Each author declares that he or she has no commercial associations that might pose a conflict of interest in connection with the submitted article

Authors' Contribution: RK: study conception and design; YS, YB: data collection and analysis; KhK: interpretation of data; RK, YS: manuscript drafting; RK, KhK: critical revision. All authors approved the final manuscript

Declaration on the Use of AI: ChatGPT (OpenAI) was used only for language editing. The authors take full responsibility for the content of the manuscript.

Consent for Publication: Not required

Acknowledgments: None

Funding: None

References

- World Obesity Atlas 2025 | World Obesity Federation [Internet]. [cited 2025 Nov 23]. Available from: <https://www.worldobesity.org/resources/resource-library/world-obesity-atlas-2025>
- Powell-Wiley TM, Poirier P, Burke LE, et al. Obesity and Cardiovascular Disease A Scientific Statement From the American Heart Association. Vol. 143, Circulation. 2021. doi:10.1161/CIR.0000000000000973
- Lam BCC, Lim AYL, Chan SL, Yum MPS, Koh NSY, Finkelstein EA. The impact of obesity: A narrative review. Vol. 64, Singapore Medical Journal. 2023. doi:10.4103/singaporemedj.SMJ-2022-232
- Pati S, Irfan W, Jameel A, Ahmed S, Shahid RK. Obesity and Cancer: A Current Overview of Epidemiology, Pathogenesis, Outcomes, and Management. Vol. 15, Cancers. 2023. doi:10.3390/cancers15020485
- Su X, Peng D. Emerging functions of adipokines in linking the development of obesity and cardiovascular diseases. Mol Biol Rep. 2020 Oct 1;47(10):7991–8006. doi:10.1007/s11033-020-05732-9
- Kim JA, Choi KM. Newly Discovered Adipokines: Pathophysiological Link Between Obesity and Cardiometabolic Disorders. Front Physiol. 2020 Sep 2;11. doi:10.3389/fphys.2020.568800
- Kojta I, Chacińska M, Błażnio-Zabielska A. Obesity, bioactive lipids, and adipose tissue inflammation in insulin resistance. Nutrients. 2020 May 1;12(5). doi:10.3390/nu12051305
- Aleksandrova K, Egea Rodrigues C, Floegel A, Ahrens W. Omics Biomarkers in Obesity: Novel Etiological Insights and Targets for Precision Prevention. Curr Obes Reports 2020 93. 2020 Jun 27;9(3):219–30. doi:10.1007/S13679-020-00393-Y
- Torretta E, Barbacini P, Al-Daghri NM, Gelfi C. Sphingolipids in obesity and correlated comorbidities: The contribution of gender, age and environment. Int J Mol Sci. 2019 Dec 1;20(23). doi:10.3390/ijms20235901
- Kucuk S, Niven J, Caamano J, et al. Unwrapping the mechanisms of ceramide and fatty acid-initiated signals leading to immune-inflammatory responses in obesity. Int J Biochem Cell Biol. 2021 Jun 1;135:105972. doi:10.1016/J.BIOCEL.2021.105972
- Hady HR, Błażnio-Zabielska AU, Szczerbiński Ł, et al. Ceramide content in liver increases along with insulin resistance in obese patients. J Clin Med. 2019 Dec 1;8(12). doi:10.3390/jcm8122197
- Yu B, Hu M, Jiang W, et al. Ceramide d18:1/24:1 as a potential biomarker to differentiate obesity subtypes with unfavorable health outcomes. Lipids Health Dis. 2023 Dec 1;22(1). doi:10.1186/s12944-023-01921-0
- Lytle KA, Chung JO, Bush NC, Triay JM, Jensen MD. Ceramide concentrations in liver, plasma, and very low-density lipoproteins of humans with severe obesity. Lipids. 2023 May 1;58(3):107–15. doi:10.1002/lipid.12367
- Kim E, Jeon S. The Impact of Phytochemicals in Obesity-Related Metabolic Diseases: Focus on Ceramide Metabolism. Nutrients. 2023 Feb 1;15(3). doi:10.3390/nu15030703
- Fucho R, Casals N, Serra D, Herrero L. Ceramides and mitochondrial fatty acid oxidation in obesity. FASEB J. 2017 Apr 1;31(4):1263–72. doi:10.1096/fj.201601156R
- Pascoa TC, Pike ACW, Tautermann CS, et al. Structural basis of the mechanism and inhibition of a human ceramide synthase. Nat Struct Mol Biol. 2025 Mar 1;32(3):431–40. doi:10.1038/s41594-024-01414-3
- Hammerschmidt P, Brüning JC. Contribution of specific ceramides to obesity-associated metabolic diseases. Cell Mol Life Sci. 2022 Aug 1;79(8). doi:10.1007/s00018-022-04401-3
- Chathoth S, Ismail MH, Alghamdi HM, et al. Insulin resistance induced by de novo pathway-generated C16-ceramide is associated with type 2 diabetes in an obese population. Lipids Health Dis. 2022 Dec 1;21(1). doi:10.1186/s12944-022-01634-w
- Maldonado-Hernández J, Saldaña-Dávila GE, Piña-Aguero MI, Núñez-García BA, López-Alarcón MG. Association between plasmatic ceramides profile and AST/ALT Ratio: C14:0 Ceramide as predictor of hepatic steatosis in adolescents independently of obesity. Can J Gastroenterol Hepatol. 2017;2017. doi:10.1155/2017/3689375
- Kayser BD, Prifti E, Lhomme M, et al. Elevated serum ceramides are linked with obesity-associated gut dysbiosis and impaired glucose metabolism. Metabolomics. 2019 Nov 1;15(11). doi:10.1007/s11306-019-1596-0
- Haus JM, Kashyap SR, Kasumov T, et al. Plasma Ceramides Are Elevated in Obese Subjects With Type 2 Diabetes and Correlate With the Severity of Insulin Resistance. Diabetes. 2009 Feb 1;58(2):337–43. doi:10.2337/DB08-1228
- Adams JM, Pratipanawatr T, Berria R, et al. Ceramide Content Is Increased in Skeletal Muscle From Obese Insulin-Resistant Humans. Diabetes. 2004 Jan 1;53(1):25–31. doi:10.2337/DIABETES.53.1.25

23. Amati F, Dubé JJ, Alvarez-Carnero E, al. Skeletal Muscle Triglycerides, Diacylglycerols, and Ceramides in Insulin Resistance: Another Paradox in Endurance-Trained Athletes? *Diabetes*. 2011 Oct 1;60(10):2588–97. doi:10.2337/db10-1221
24. Chaurasia B, Summers SA. Ceramides – Lipotoxic Inducers of Metabolic Disorders. *Trends Endocrinol Metab*. 2015 Oct 1;26(10):538–50. doi:10.1016/j.tem.2015.07.006
25. Huang Y, Sulek K, Stinson SE, et al. Lipid profiling identifies modifiable signatures of cardiometabolic risk in children and adolescents with obesity. *Nat Med* 2024 311. 2024 Sep 20;31(1):294–305. doi:10.1038/s41591-024-03279-x
26. Hammerschmidt P, Brüning JC. Contribution of specific ceramides to obesity-associated metabolic diseases. *Cell Mol Life Sci* 2022 798. 2022 Jul 5;79(8):395–. doi:10.1007/s00018-022-04401-3
27. Kosyakov D, Pisyakov V. “I’d like to publish in Q1, but there’s no Q1 to be found”: Study of journal quartile distributions across subject categories and topics. *J Informetr*. 2024 Feb 1;18(1):101494. doi:10.1016/j.joi.2024.101494
28. Ballesteros-Ballesteros VA, Zárate-Torres RA. Mapping the conceptual structure of research on open innovation in university–industry collaborations: a bibliometric analysis. *Front Res Metrics Anal*. 2025 Nov 28;10:1693969. doi:10.3389/frma.2025.1693969
29. Ou G, Zhao K, Zuo R, Wu J. Effects of research funding on the academic impact and societal visibility of scientific research. *J Informetr*. 2024 Nov 1;18(4):101592. doi:10.1016/j.joi.2024.101592
30. McManus C, Neves AAB, Filho JAD, Pimentel F, Pimentel D. Funding as a determinant of Citation Impact in Scientific Papers in different countries. *An Acad Bras Cienc*. 2023;95(1):e20220515. doi:10.1590/0001-3765202320220515
31. Wang J, Frietsch R, Neuhausler P, Hooi R. International collaboration leading to high citations: Global impact or home country effect? *J Informetr*. 2024 Nov 1;18(4):101565. doi:10.1016/j.joi.2024.101565
32. Diprete TA, Eirich GM. Cumulative advantage as a mechanism for inequality: A review of theoretical and empirical developments. *Annu Rev Sociol*. 2006;32:271–97. doi:10.1146/annurev.soc.32.061604.123127
33. Vieira ES. The influence of research collaboration on citation impact: the countries in the European Innovation Scoreboard. *Sci* 2023 1286. 2023 May 8;128(6):3555–79. doi:10.1007/s11192-023-04715-4
34. Franceschet M, Costantini A. The effect of scholar collaboration on impact and quality of academic papers. *J Informetr*. 2010 Oct 1;4(4):540–53. doi:10.1016/j.joi.2010.06.003
35. Meikle TG, Huynh K, Giles C, Meikle PJ. Clinical lipidomics: realizing the potential of lipid profiling. *J Lipid Res*. 2021 Jan 1;63:100127. doi:10.1016/J.JLR.2021.100127
36. Duarte I, Santos HGG, Rodrigues ML. A general analysis of the impact of international collaboration on the citation indices of scientific publications from 60 institutions across five continents. *An Acad Bras Cienc*. 2025;97(1):e20241035. doi:10.1590/0001-3765202520241035
37. Delcheva G, Stefanova K, Stankova T. Ceramides—Emerging Biomarkers of Lipotoxicity in Obesity, Diabetes, Cardiovascular Diseases, and Inflammation. *Dis* 2024, Vol 12,. 2024 Aug 23;12(9). doi:10.3390/diseases12090195
38. And H, Sciences Communications | S. Higher education efficiency and productivity studies: a bibliometric analysis for future research. *Humanit Soc Sci Commun* 2026 131. 2026 Jan 14;13(1):100–. doi:10.1057/s41599-025-06171-2
39. Beyene HB, Giles C, Huynh K, et al. Metabolic phenotyping of BMI to characterize cardiometabolic risk: evidence from large population-based cohorts. *Nat Commun* 2023 141. 2023 Oct 7;14(1):6280–. doi:10.1038/s41467-023-41963-7
40. Hennemann S, Rybski D, Liefner I. The myth of global science collaboration—Collaboration patterns in epistemic communities. *J Informetr*. 2012 Apr 13;6(2):217–25. doi:10.1016/j.joi.2011.12.002
41. Nast C, Llopi O, Yankova D, D’Este P. Sourcing insights elsewhere: The positive influence of academic engagement on scientific impact. *Technovation*. 2025 Jan 1;139(2):103112. doi:10.1016/j.technovation.2024.103112
42. Roden M, Price TB, Perseghin G, et al. Mechanism of free fatty acid-induced insulin resistance in humans. *J Clin Invest*. 1996 Jun 15;97(12):2859–65. doi:10.1172/JCI118742
43. Itani SI, Ruderman NB, Schmieder F, Boden G. Lipid-Induced Insulin Resistance in Human Muscle Is Associated With Changes in Diacylglycerol, Protein Kinase C, and IκB-α. *Diabetes*. 2002 Jul 1;51(7):2005–11. doi:10.2337/diabetes.51.7.2005
44. Shulman GI. Cellular mechanisms of insulin resistance. *J Clin Invest*. 2000 Jan 15;106(2):171–6. doi:10.1172/JCI10583
45. Quehenberger O, Armando AM, Brown AH, et al. Lipidomics reveals a remarkable diversity of lipids in human plasma. *J Lipid Res*. 2010 Nov 1;51(11):3299–305. doi:10.1194/jlr.M009449
46. Havulinna AS, Sysi-Aho M, Hilvo M, al. Circulating Ceramides Predict Cardiovascular Outcomes in the Population-Based FINRISK 2002 Cohort. *Arterioscler Thromb Vasc Biol*. 2016 Dec 1;36(12):2424–30. doi:10.1161/ATVBAHA.116.307497;CTYPE:STRING:JOURNAL

Copyright: The Author(s), 2026. Licensee Mattioli 1885, Fidenza, Italy. This is an open-access article distributed under the terms of the Creative Commons Attribution NonCommercial License (CC BY-NC-4.0).

Disclaimer/Publisher’s Note: The statements, opinions and data contained in this article are solely those of the author(s) and contributor(s) and do not necessarily reflect those of their affiliated organizations, the publisher, the editors or the reviewers. The publisher and the editors disclaim any responsibility for injury to people or property resulting from any ideas, methods, instructions or products mentioned in the content. Any product that may be evaluated in this article, or claim made by its manufacturer, is not guaranteed or endorsed by the publisher.